

Bypass firewalls with of-CORs and typo-squatting ♦ Truffle Security Co.

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Content

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Chris Grayson

January 3, 2023

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CORS can be configured in two different ways. The first is to specify an origin that can read responses. In this case, logged in users can have their user data accessed. The second is to use a `**wildcard`, `**`which will not send login session information and is usually assumed to be safe for that reason. That's often true for externally facing websites but can go horrendously wrong for internal facing web apps that don't use authentication.

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Origin: some.other.origin.com
Accept-Language: en-us
Accept-Encoding: gzip, deflate
Connection: Keep-Alive

HTTP/1.1 200 OK
Date: Mon, 12 Dec 2022 00:00:00 GMT
Server: Some Cool Server
Content-Length: 230
Access-Control-Allow-Origin: *
Content-Type: text/html; charset=iso-8859-1
Connection: Closed
```

[PortSwigger](#) has done an excellent job explaining this and talks through these misconfigurations in much more detail. For this post we will just stay focused on the issue of using wildcards for internal apps.

You'll find such an example of an incomplete view of wildcard CORS on websites like <https://enable-cors.org/> which fails to mention one of the most dangerous aspects of enabling CORS; its use on internal networks. People's browsers straddle multiple networks so when a victim visits an evil website that evil website can hit all of the internal apps on internal networks.

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The core hypothesis of-CORS was built to test was "large internal corporate networks are exceedingly likely to have impactful CORS misconfigurations." As such we wanted a toolkit that would do all the following:

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Utilize a service worker to make requests long after the victim is redirected off the typosquatting domain

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Achieving these goals took a bit of work in a handful of domains.

The Web Application

The of-CORS web application is a Python3 application built using Django and Django Rest Framework. When a victim visits the web application a lookup is done to determine what internal domains are configured to be probed. A browser service worker is then registered to do the probing and the user's browser is quickly redirected away to the *assumed* intended destination.

For example, let's say that we have registered eslamotors.com and pointed it to our of-CORS instance. An internal Tesla employee then accidentally visits <https://foobar.eslamotors.com/bing/bang/bong.php> (whoops, typo alert!). of-CORS then looks up the core domain of "eslamotors.com" to determine the list of domains that a payload should be launched for. This JavaScript payload for running the probing is then registered as a service worker and the employee's browser is redirected to <https://foobar.teslamotors.com/bing/bang/bong.php>. The service worker continues to run in the background, issuing HTTPS requests to all of the configured domains and reporting back to of-CORS any identified CORS misconfigurations *as well as* the HTML content for the misconfigured domains.

Just like that, we have a setup for probing internal network CORS misconfigurations with minimal indication to the victim. Any internal apps without authentication and permissive CORS will send all their data to your instance of of-CORS.

The Infrastructure

We wanted a toolkit that enabled us to receive HTTPS requests for a myriad of different domains. We also wanted the ability to spin new domains and targets up and down with relative ease. Thus we needed someone else to handle the SSL/TLS certificate management.

The deployment we landed on was using Cloudflare for SSL/TLS termination and DNS management and Heroku for application hosting. Cloudflare allows for wildcard CNAME routing and Heroku allows for configuring arbitrary domains to point to existing Heroku stacks. Together they achieve the infrastructure flexibility we needed for rapid iteration.

Even with this relatively simple deployment, managing infrastructure can be a real pain. To address this we implemented Terraform configuration that wires up Cloudflare, Heroku, and of-CORS in the correct configuration based on a single YAML file.

Getting Victims to Visit Our Application

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Human error and bad typing to the rescue! We decided to go the route of purchasing [typo-squat](#) domains that were very similar to the internal domains used by the organizations we targeted. You'll need to do a little reconnaissance to learn what internal second level domains your target company uses. You can often find this in old commits in Github repos, android builds, and sometimes even StackOverflow questions. An example of an internal Uber domain found via GitHub is shown below:

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Configuring the Stack

So we've got the application, the sneaky domains to get victims to come say hi, and the infrastructure. Now we just need the data! The of-CORS configuration file not only specifies the infrastructure setup, but also which payloads should be generated for which domains. Take a look at the following sample configuration file:

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terraform:
# You must change this to a unique string that is a valid Heroku app name
heroku_app_name: best-cors-hunter
# Fill this out with your Cloudflare API token
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hosts:
  testing:
    host_domain: 127.0.0.1:8080
    redirect_domain: google.com
    targets:
      - enable-cors.org
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Under the hosts.testing configuration directive we see a host_domain value of 127.0.0.1:8080, a redirect_value domain of google.com, and two targets configured for enable-cors.org and example.com. In this configuration of-CORS will expect to receive requests at 127.0.0.1:8080, will subsequently launch a payload targeting subdomains of example.com and enable-cors.org when a request is received, and will redirect the victim's browser to google.com after the browser service worker is registered.

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#Secret Scanning

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